Selenium Command Mastery – The Ultimate Reference Guide

1. Browser Commands

- Control browser session, launch URLs, and manage browser windows.

FAQ:

Q: How do you launch a browser and navigate to a URL?
A: Use driver.get("url").

Q: What's the difference between driver.close() and driver.quit()?
A: close() closes current tab; quit() ends the session.

Q: How to get page title or URL?
A: Use driver.getTitle() and driver.getCurrentUrl().

```
driver.get("https://example.com");
driver.getTitle();
driver.getCurrentUrl();
driver.close();
driver.quit();
```

2. Navigation Commands

- Move forward, backward, or refresh browser history and URLs.

FAQ:

Q: How do you navigate back or forward?
A: Use driver.navigate().back() or .forward().

Q: How to refresh a page?
A: Use driver.navigate().refresh().

Q: How is get() different from navigate().to()?
A: navigate().to() adds to history stack.

```
driver.navigate().to("https://example.com");
driver.navigate().back();
driver.navigate().forward();
driver.navigate().refresh();
```

3. Window Management Commands

- Resize, reposition, and manage multiple windows or tabs.

FAQ:

Q: How to maximize or minimize a window?
A: Use window().maximize() or .minimize().

Q: How to switch tabs?
A: Use windowHandles and switchTo().window(handle).

Q: How to get window size?
A: Use driver.manage().window().getSize();

```
driver.manage().window().maximize();
driver.manage().window().setSize(new Dimension(1024, 768));
String mainHandle = driver.getWindowHandle();
Set allHandles = driver.getWindowHandles();
driver.switchTo().window(mainHandle);
```

4. Frame and IFrame Commands

- Switch between frames and iframes to interact with embedded elements.

FAQ:
Q: Why do tests fail inside iframes?
A: You must switch to the correct frame first.

Q: How to switch to default content?
A: Use driver.switchTo().defaultContent();

Q: How to handle nested frames?
A: Switch to parent frame first, then child.

```
driver.switchTo().frame("frameName");
driver.switchTo().defaultContent();
driver.switchTo().parentFrame();
```

5. Alert Commands

- Handle alerts, confirms, and prompt boxes.

FAQ:
Q: How to accept an alert?
A: Use driver.switchTo().alert().accept();

Q: How to dismiss an alert?
A: Use .dismiss();

Q: How to get alert text?
A: Use .getText();

```
Alert alert = driver.switchTo().alert();
alert.accept();
alert.dismiss();
String alertText = alert.getText();
```

6. Element Interaction Commands

- Find elements using various locator strategies.

FAQ:

Q: How to find an element by ID?
A: Use driver.findElement(By.id("id"));

Q: How to find multiple elements?
A: Use findElements();

Q: How to locate by XPath?
A: Use By.xpath("expression");

```
driver.findElement(By.id("username"));
driver.findElements(By.className("input-class"));
driver.findElement(By.xpath("//input[@type='text']"));
```

7. Element Action Commands

- Click, type, clear, and submit input fields or buttons.

FAQ:
Q: How to click a button?
A: Use .click();

Q: How to type into a field?
A: Use .sendKeys("text");

Q: How to clear an input?
A: Use .clear();

```
driver.findElement(By.id("submit")).click();
driver.findElement(By.id("email")).sendKeys("test@example.com");
driver.findElement(By.id("email")).clear();
```

8. Wait Commands

- Synchronize tests by waiting for elements or conditions.

FAQ:
Q: What is an implicit wait?
A: Global timeout for element searches.

Q: What is explicit wait?
A: Wait for specific condition.

Q: Why not mix both?
A: It causes timeout conflicts.

```
driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("element")));
```

9. Keyboard & Mouse Action Commands

- Simulate complex user gestures using Actions class.

FAQ:

Q: How to hover or double-click?

A: Use moveToElement(), doubleClick().

Q: How to right-click?

A: Use contextClick().

Q: How to drag and drop?

A: Use dragAndDrop(source, target).

```
Actions actions = new Actions(driver);
actions.moveToElement(element).perform();
actions.doubleClick(element).perform();
actions.contextClick(element).perform();
```

10. Cookies Management Commands

- Add, retrieve, or delete cookies during a session.

FAQ:

Q: How to add a cookie?

A: Use addCookie(new Cookie(...));

Q: How to delete all cookies?

A: Use deleteAllCookies();

Q: How to read cookies?

A: Use getCookies();

```
driver.manage().addCookie(new Cookie("test", "value"));
driver.manage().getCookies();
driver.manage().deleteAllCookies();
```

11. Screen Capture Commands

- Take screenshots for debugging or reporting.

FAQ:

Q: How to take a screenshot?

A: Use getScreenshotAs()

Q: Where to save the image?

A: Use FileUtils.copyFile().

Q: Capture as Base64?

A: Use OutputType.BASE64

```
File src = ((TakesScreenshot)driver).getScreenshotAs(OutputType.FILE);
FileUtils.copyFile(src, new File("screenshot.png"));
```

12. Session Management Commands

- Control the lifetime of browser sessions.

FAQ:

Q: How to quit session?

A: Use driver.quit();

Q: How to close current tab?

A: Use driver.close();

Q: How to reset cookies?

A: Use deleteAllCookies().

```
driver.quit();
driver.close();
driver.manage().deleteAllCookies();
```

13. Browser Options & Capabilities

- Customize browser behavior before launch.

FAQ:

Q: How to run Chrome headless?

A: Add 'headless' to ChromeOptions.

Q: How to disable extensions?

A: Add 'disable-extensions' flag.

Q: What are capabilities?

A: Settings to configure WebDriver.

```
ChromeOptions options = new ChromeOptions();
options.addArguments("headless", "disable-extensions");
WebDriver driver = new ChromeDriver(options);
```

14. Logs Management Commands

- Access browser or driver logs.

FAQ:

Q: How to get logs?

A: Use driver.manage().logs().get("type");

Q: What types exist?

A: browser, driver, client, server.

Q: Use cases?

A: Debugging JS errors, network issues.

```
driver.manage().logs().get("browser");
driver.manage().logs().get("driver");
```

15. JavaScript Execution Commands

- Execute JS directly in the browser.

FAQ:

Q: How to scroll with JS?

A: Use window.scrollTo().

Q: Click using JS?

A: Use arguments[0].click();

Q: Why JS?

A: For unsupported actions.

```
JavascriptExecutor js = (JavascriptExecutor) driver;
js.executeScript("window.scrollTo(0, 500);");
js.executeScript("arguments[0].click();", element);
```

16. File Upload and Download Commands

- Handle file selection and triggering downloads.

FAQ:

Q: How to upload?

A: Use sendKeys(path).

Q: How to download?

A: Click a link or button.

Q: Dialog handling?

A: Needs OS tool, not Selenium.

```
driver.findElement(By.id("upload")).sendKeys("/path/to/file.txt");
driver.findElement(By.linkText("Download")).click();
```

17. Proxy Management Commands

- Configure WebDriver to use a proxy.

FAQ:

Q: How to use proxy?

A: Set in DesiredCapabilities or Options.

Q: When to use?

A: Network filtering, geo-location.

Q: Tools to monitor?

A: Use BrowserMob Proxy.

```
Proxy proxy = new Proxy();
proxy.setHttpProxy("localhost:8080");
ChromeOptions options = new ChromeOptions();
options.setCapability("proxy", proxy);
```

18. Mobile Web Testing Commands

- Run tests on mobile browsers via Appium.

FAQ:

Q: Use same locators?

A: Yes, supported.

Q: Use gestures?

A: Supported via TouchActions.

Q: Setup required?

A: Install Appium + device config.

```
driver.findElement(By.id("search")).click();
driver.findElement(By.id("search")).sendKeys("Test");
```

19. Remote WebDriver Commands

- Run tests on remote grid or cloud services.

FAQ:

Q: How to connect remotely?

A: Use RemoteWebDriver + URL.

Q: When to use?

A: Parallel, distributed testing.

Q: Supported providers?

A: Selenium Grid, BrowserStack, SauceLabs.

```
WebDriver driver = new RemoteWebDriver(new URL("http://localhost:4444/wd/hub"),
capabilities);
```

20. WebDriver Timeout Commands

- Set limits on script, page, and element waits.

FAQ:

Q: How to set page load timeout?

A: Use pageLoadTimeout().

Q: Script timeout?

A: Use setScriptTimeout().

Q: Element wait?

A: Use implicitlyWait().

```
driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
driver.manage().timeouts().pageLoadTimeout(Duration.ofSeconds(30));
driver.manage().timeouts().setScriptTimeout(Duration.ofSeconds(30));
```